

International Economic Relations

ECON 308

The Standard Trade Model
(Chapter 6)

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Introduction (1 of 3)

- **Standard trade model** is a general model that includes Ricardian, specific factors, and Heckscher-Ohlin models as special cases.
 - Two goods, food (F) and cloth (C).
 - Each country's PPF is a smooth curve.

Introduction (2 of 3)

- Differences in **labor services, labor skills, physical capital, land, and technology** between countries cause differences in production possibility frontiers.
- A country's PPF determines its relative supply function.
- National relative supply functions determine a world relative supply function, which along with world relative demand determines the equilibrium under international trade.

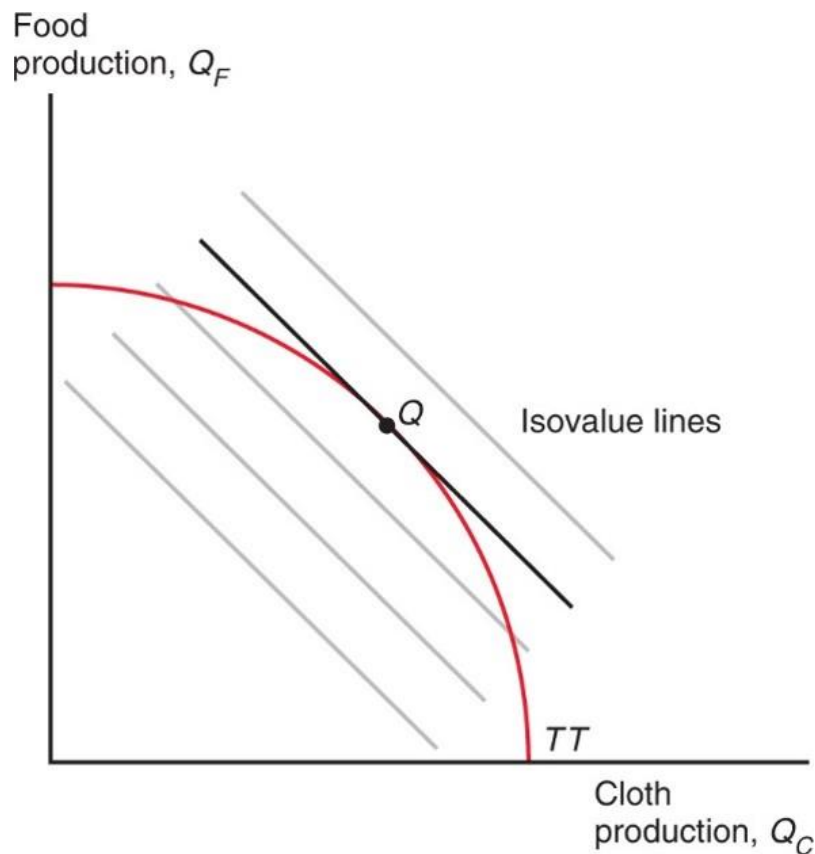
Introduction (3 of 3)

- The standard trade model is built on four key relationships:
 1. the relationship between the production possibility frontier and the relative supply curve,
 2. the relationship between relative prices and relative demand,
 3. the determination of world equilibrium by world relative supply and world relative demand, and
 4. the effect of the terms of trade—the price of a country's exports divided by the price of its imports—on a nation's welfare.

Production Possibilities and Relative Supply (1 of 2)

- What a country produces depends on the relative price of cloth to food $\frac{P_C}{P_F}$.
- An economy chooses its production of cloth Q_C and food Q_F to maximize the value of its output $V = P_C Q_C + P_F Q_F$, given the prices of cloth and food.
- The slope of an isovalue line equals $-\left(\frac{P_C}{P_F}\right)$.
- Produce at point where PPF is tangent to isovalue line.

Figure 6.1 Relative Prices Determine the Economy's Output

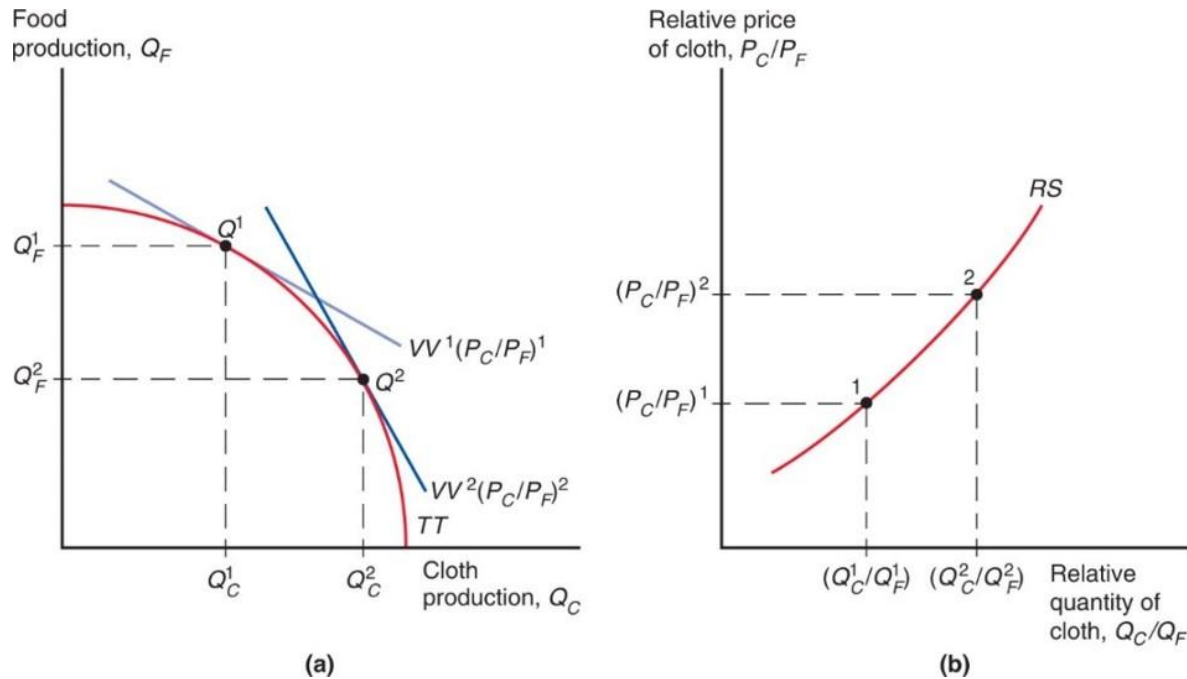


An economy whose production possibility frontier is TT will produce at Q , which is on the highest possible isovalue line.

Production Possibilities and Relative Supply (2 of 2)

- Relative prices and relative supply:
 - An increase in the price of cloth relative to food $\frac{P_C}{P_F}$ makes the isovalue line steeper.
 - Production shifts from point Q^1 to point Q^2
 - Supply of cloth relative to food $\frac{Q_C}{Q_F}$ rises.
 - Relative supply of cloth to food increases with the relative price of cloth to food.

Figure 6.2 How an Increase in the Relative Price of Cloth Affects Relative Supply



In panel (a), the isovalue lines become steeper when the relative price of cloth rises. As a result, the economy produces more cloth and less food. Panel (b) shows the relative supply curve associated with the production possibilities frontier TT . The rise in the relative price of cloth leads to an increase in the relative production of cloth.

Relative Prices and Demand (1 of 6)

- The value of the economy's consumption must equal the value of the economy's production.

$$P_C D_C + P_F D_F = P_C Q_C + P_F Q_F = V$$

- Assume that the economy's consumption decisions may be represented as if they were based on the tastes of a single representative consumer.
- An **indifference curve** represents combinations of cloth and food that leave the consumer equally well off (indifferent).

Relative Prices and Demand (2 of 6)

- Indifference curves
 - are downward sloping—if you have less cloth, then you must have more food to be equally satisfied.
 - that lie farther from the origin make consumers more satisfied—they prefer having more of both goods.
 - become flatter when they move to the right—with more cloth and less food, an extra yard of cloth becomes less valuable in terms of how many calories of food you are willing to give up for it.

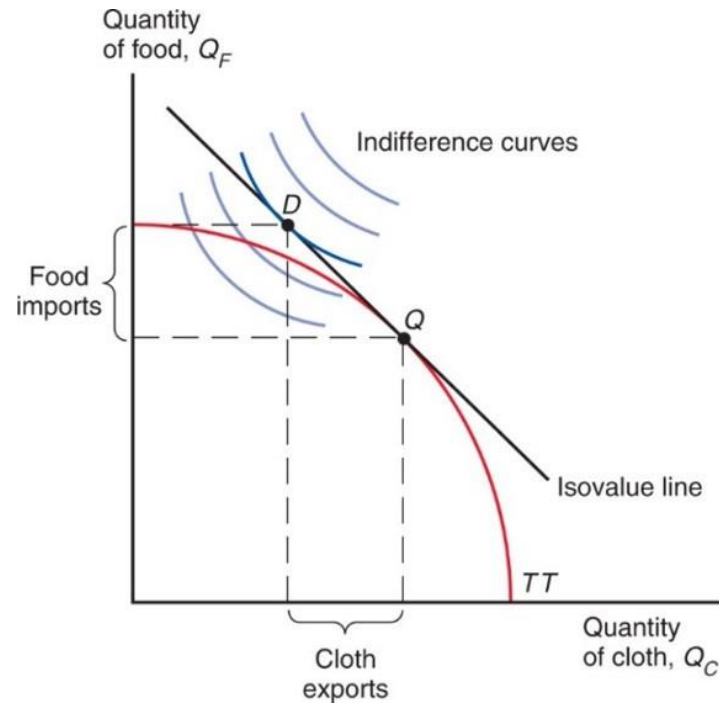
Relative Prices and Demand (3 of 6)

- Consumption choice is based on preferences and relative price of goods:
 - Consume at point D where the isovalue line is tangent to the indifference curve.
- Economy exports cloth—the quantity of cloth produced exceeds the quantity of cloth consumed—and imports food.

Relative Prices and Demand (4 of 6)

- Relative prices and relative supply:
 - An increase in the relative price of cloth $\frac{P_C}{P_F}$ causes consumption choice to shift from point D^1 to point D^2
 - Demand for cloth relative to food $\frac{D_C}{D_F}$ falls.
 - Relative supply of cloth to food increases with the relative price of cloth to food.

Figure 6.3 Production, Consumption, and Trade in the Standard Model



The economy produces at point Q , where the production possibility frontier is tangent to the highest possible isovalue line. It consumes at point D , where that isovalue line is tangent to the highest possible indifference curve. The economy produces more cloth than it consumes and therefore exports cloth; correspondingly, it consumes more food than it produces and therefore imports food.

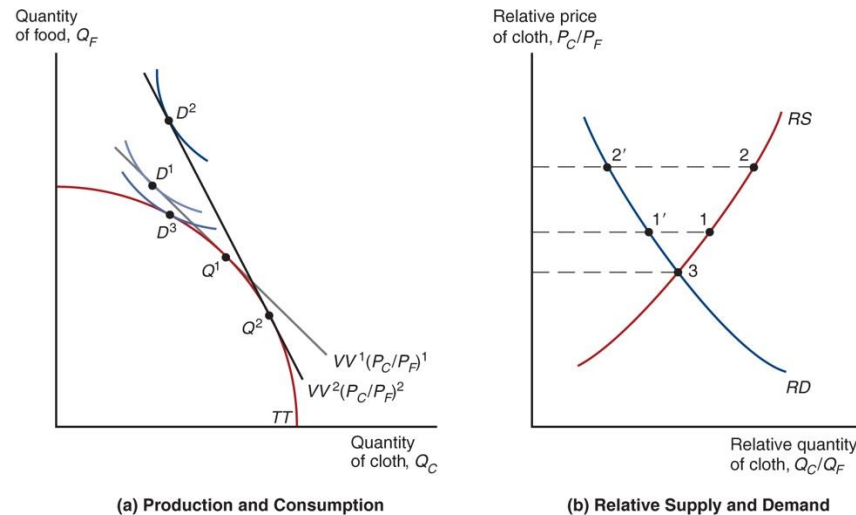
Relative Prices and Demand (5 of 6)

- An economy that exports cloth is better off when the price of cloth rises relative to the price of food:
 - the isovalue line becomes steeper and a higher indifference curve can be reached.
- A higher relative price of cloth means that more calories of food can be imported for every yard of cloth exported.

Relative Prices and Demand (6 of 6)

- If the economy cannot trade:
 - The relative price of cloth to food is determined by the intersection of relative demand and relative supply for that country.
 - Consume and produce at point D^3 where the indifference curve is tangent to the production possibilities frontier.

Figure 6.4 Effects of a Rise in the Relative Price of Cloth and Gains from Trade



In panel (a), the slope of the isoquant lines is equal to minus the relative price of cloth. As a result, when that relative price rises, all isoquant lines become steeper. Production shifts from Q_1 to Q_2 and consumption shifts from D_1 to D_2 . If the economy cannot trade, then it produces and consumes at point D_3 .

Panel (b) shows the effects of the rise in the relative price of cloth on relative production (move from 1 to 2) and relative demand (move from 1 to 2'). If the economy cannot trade, then it consumes and produces at point 3.

U.S. Consumer Gains from Chinese Imports (1 of 2)

- The aggregate gains from trade are in large part driven by consumers' access to cheaper imported goods.
- Increased trade between the United States and China has increased the purchasing power of U.S. consumers.
- A recent study finds that increased trade with China between 2000 and 2007 induced large reductions in prices, generating substantial increases in the purchasing power of U.S. households.
- On average those gains represented \$1,500 per household per year.

U.S. Consumer Gains from Chinese Imports (2 of 2)

- Over half of those gains were generated by price reductions of U.S. produced goods in response to the increased competition from Chinese imports.
- And because those price decreases were more pronounced for goods consumed by relatively poorer households with annual incomes below \$30,000, the average \$1,500 gain was 36 percent higher for those households relative to richer households with income above \$100,000.

The Welfare Effects of Changes in the Terms of Trade

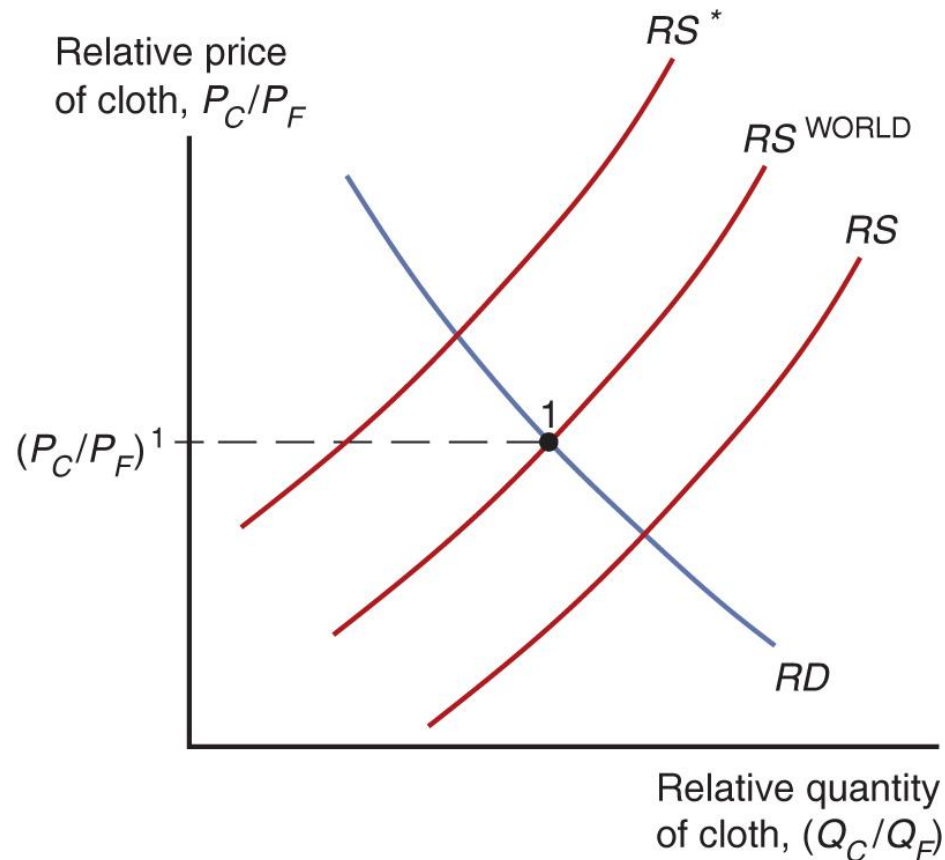
- The **terms of trade** refers to the price of exports relative to the price of imports.
 - When a country exports cloth and the relative price of cloth increases, the terms of trade rise.
- Because a higher relative price for exports means that the country can afford to buy more imports, an increase in the terms of trade increases a country's welfare.
- A decline in the terms of trade decreases a country's welfare.

Determining Relative Prices

- To determine the price of cloth relative to the price food, use relative supply and relative demand.
 - **World** supply of cloth relative to food at each relative price.
 - **World** demand for cloth relative to food at each relative price.
- World quantities are the sum of quantities from the two countries in the world:

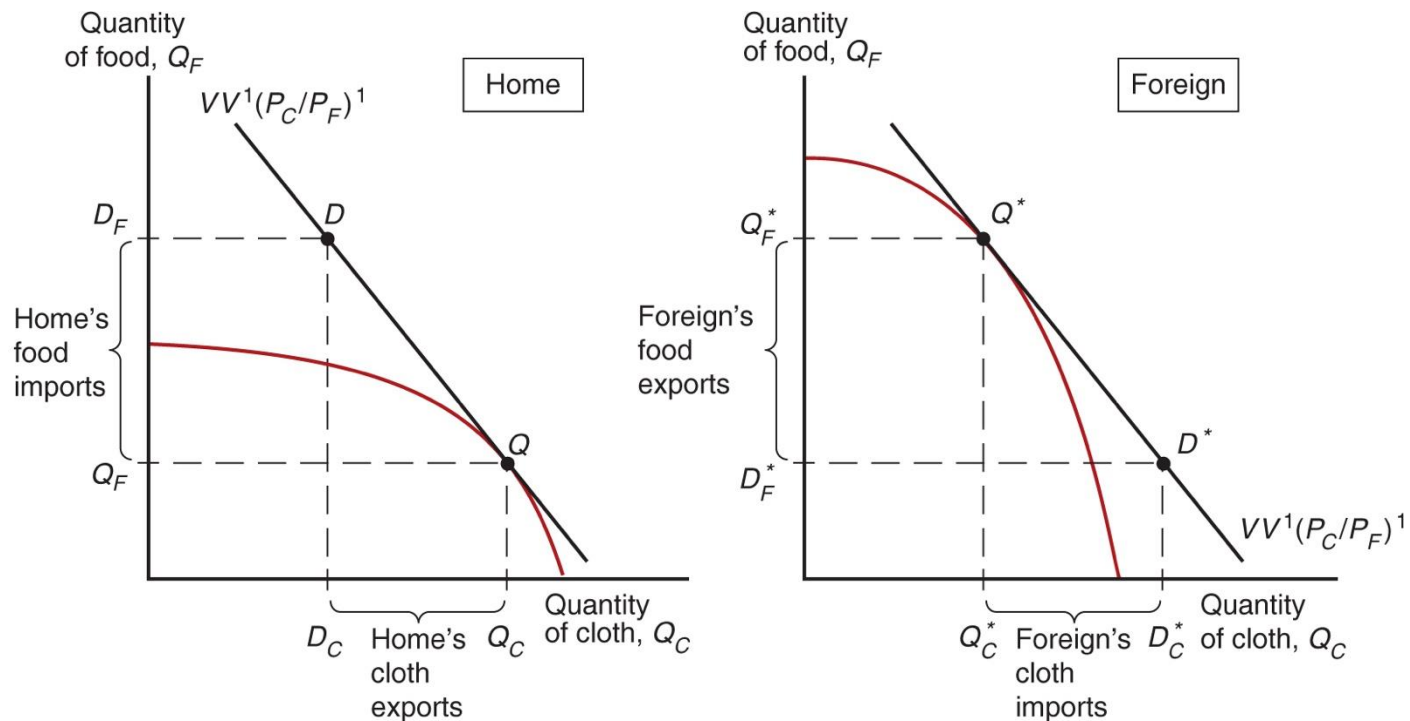
$$\frac{(Q_C + Q_C^*)}{(Q_F + Q_F^*)} \text{ and } \frac{(D_C + D_C^*)}{(D_F + D_F^*)}.$$

Figure 6.5a Equilibrium Relative Price with Trade and Associated Trade Flows



(a) Relative Supply and Demand

Figure 6.5b Equilibrium Relative Price with Trade and Associated Trade Flows



(b) Production, Consumption, and Trade

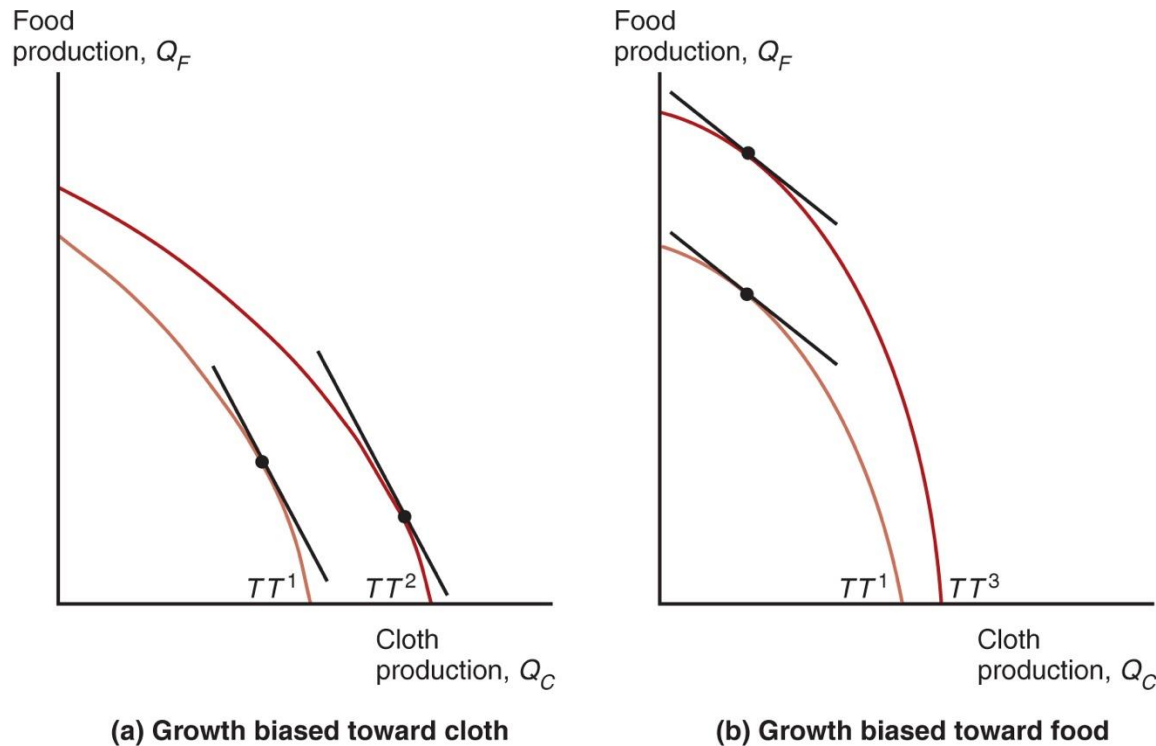
International Effects of Growth (1 of 5)

- Is economic growth in China good for the standard of living in the United States?
- Is growth in a country more or less valuable when it is integrated in the world economy?
- The standard trade model gives us precise answers to these questions.

International Effects of Growth (2 of 5)

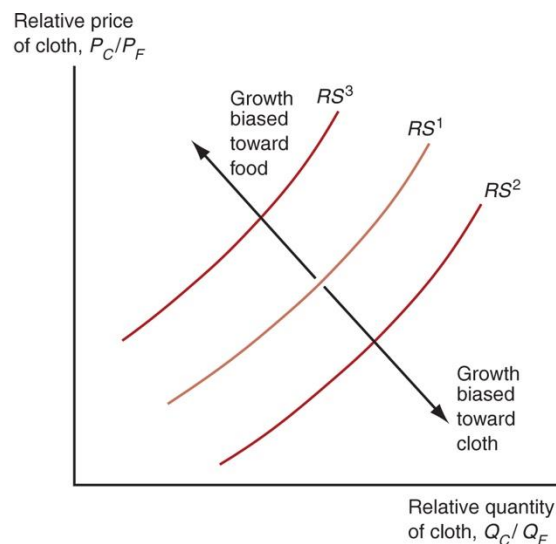
- Growth is usually **biased**: it occurs in one sector more than others, causing relative supply to change.
 - Rapid growth has occurred in U.S. computer industries but relatively little growth has occurred in U.S. textile industries.
 - In the Ricardian model, technological progress in one sector causes biased growth.
 - In the Heckscher-Ohlin model, an increase in one factor of production causes biased growth.

Figure 6.6 Biased Growth (1 of 2)



Growth is biased when it shifts production possibilities out more toward one good than toward another. In case (a), growth is biased toward cloth, while in case (b), growth is biased toward food.

Figure 6.6 Biased Growth (2 of 2)



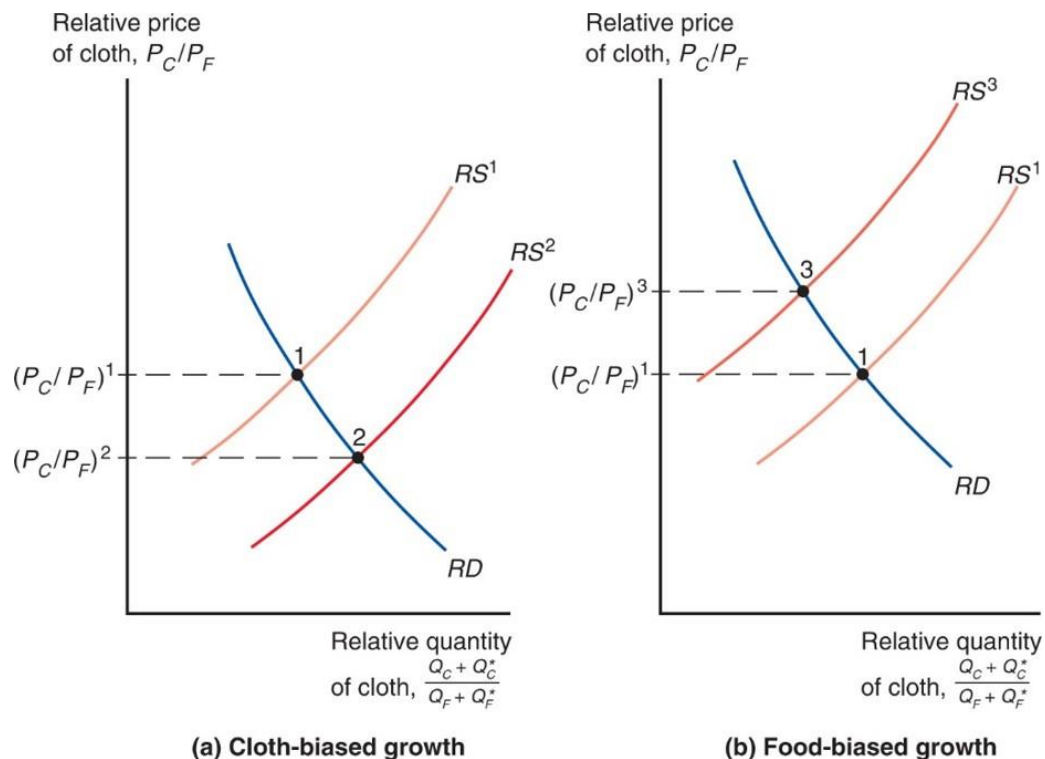
(c) Effects of biased growth on relative supply

The associated shifts in the relative supply curve are shown in panel (c): shift to the right (from RS^1 to RS^2) when growth is biased toward cloth, and shift to the left (from RS^1 to RS^3) when growth is biased toward food.

International Effects of Growth (3 of 5)

- Biased growth and the resulting change in relative supply causes a change in the terms of trade.
 - Biased growth in the cloth industry (in either the home or foreign country) will lower the price of cloth relative to the price of food and lower the terms of trade for cloth exporters.
 - Biased growth in the food industry (in either the home or foreign country) will raise the price of cloth relative to the price of food and raise the terms of trade for cloth exporters.
- Suppose that the home country exports cloth and imports food.

Figure 6.7 Growth and World Relative Supply



Growth biased toward cloth shifts the RS curve for the world to the right (a), while growth biased toward food shifts it to the left (b).

International Effects of Growth (4 of 5)

- **Export-biased growth** is growth that expands a country's production possibilities disproportionately in that country's export sector.
 - Biased growth in the food industry in the foreign country is export-biased growth for the foreign country.
- **Import-biased growth** is growth that expands a country's production possibilities disproportionately in that country's import sector.
 - Biased growth in cloth production in the foreign country is import-biased growth for the foreign country.

International Effects of Growth (5 of 5)

- Export-biased growth reduces a country's terms of trade, reducing its welfare and increasing the welfare of foreign countries.
- Import-biased growth increases a country's terms of trade, increasing its welfare and decreasing the welfare of foreign countries.

Has the Growth of Newly Industrializing Economies Hurt Advanced Nations? (1 of 3)

- Has the United States suffered from a deterioration in its terms of trade as some of its main trading partners (such as China) experienced rapid growth?
 - This would represent an aggregate income loss for the United States.
- The standard trade model predicts that if the growth in China is mainly **import-biased**, the growth in sectors that compete with U.S. exports would reduce the U.S. terms of trade.

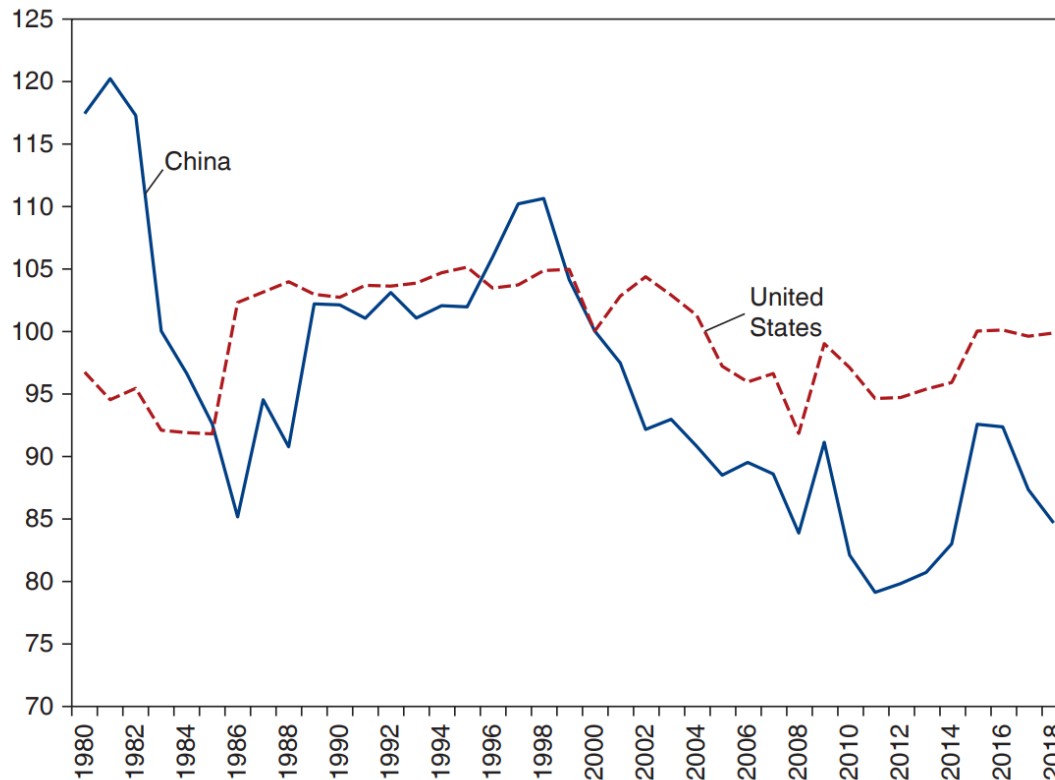
Has the Growth of Newly Industrializing Economies Hurt Advanced Nations? (2 of 3)

- But data indicates that changes in the U.S. terms of trade have been small with no clear trend over last few decades.
 - The U.S. terms of trade in 2014 was essentially at the same level as it was in 1980.
- The terms of trade for China have deteriorated over the past decade, suggesting their recent growth may have been export-biased.

Has the Growth of Newly Industrializing Economies Hurt Advanced Nations? (3 of 3)

- Like the United States, most developed countries tend to experience mild swings in their terms of trade, around 1 percent or less a year (on average).
- Some developing countries' exports are heavily concentrated in mineral and agricultural sectors.
 - The prices of those goods on world markets are very volatile, leading to large swings in the terms of trade.
 - These swings in turn translate into substantial changes in welfare.

Figure 6.8 Evolution of the Terms of Trade for the United States and China



Source: World Development Indicators, World Bank.

Import Tariffs and Export Subsidies: Simultaneous Shifts in *RS* and *RD*

- **Import tariffs** are taxes levied on imports.
- **Export subsidies** are payments given to domestic producers that export.
- Both policies influence the terms of trade and therefore national welfare.
- Import tariffs and export subsidies drive a wedge between prices in world markets and prices in domestic markets.

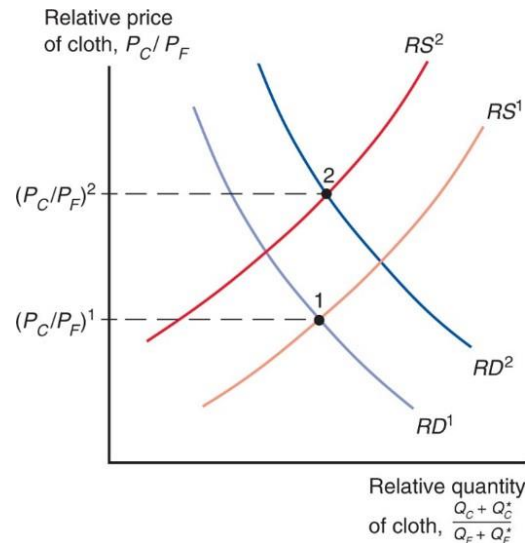
Relative Price and Supply Effects of a Tariff (1 of 2)

- If the home country imposes a tariff on food imports, the price of food relative to the price of cloth rises for domestic consumers.
 - Likewise, the price of cloth relative to the price of food falls for domestic consumers.
- Domestic producers will receive a lower relative price of cloth, and therefore will be more willing to switch to food production: relative supply of cloth will decrease.
- Domestic consumers will pay a lower relative price for cloth, and therefore will be more willing to switch to cloth consumption: relative demand for cloth will increase.

Relative Price and Supply Effects of a Tariff (2 of 2)

- When the home country imposes an import tariff, the terms of trade increase and the welfare of the country may increase.
- The magnitude of this effect depends on the size of the home country relative to the world economy.
 - If the country is a small part of the world economy, its tariff (or subsidy) policies will not have much effect on world relative supply and demand, and thus on the terms of trade.
 - But for large countries, a tariff may maximize national welfare at the expense of foreign countries.

Figure 6.9 Effects of a Food Tariff on the Terms of Trade



An import tariff on food imposed by Home both reduces the relative supply of cloth (from RS^1 to RS^2) and increases the relative demand (from RD^1 to RD^2) for the world as a whole. As a result, the relative

price of cloth must rise from $\left(\frac{P_C}{P_F}\right)^1$ to $\left(\frac{P_C}{P_F}\right)^2$.

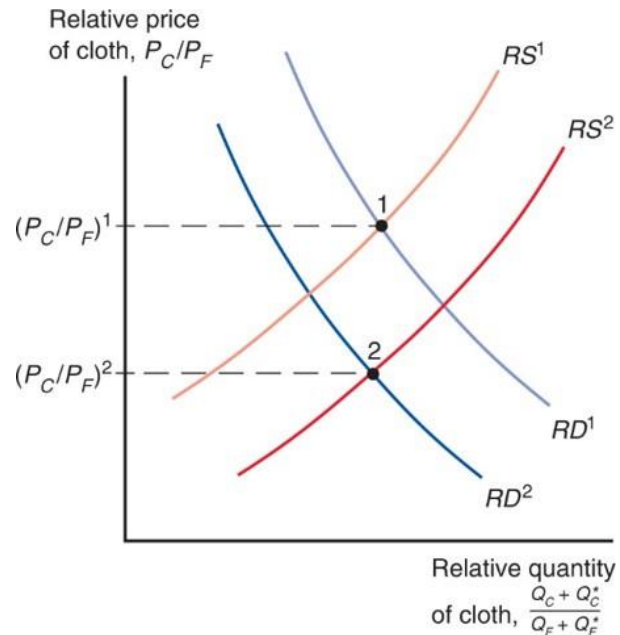
Effects of an Export Subsidy (1 of 2)

- If the home country imposes a subsidy on cloth exports, the price of cloth relative to the price of food rises for domestic consumers.
 - Domestic producers will receive a higher relative price of cloth when they export, and therefore will be more willing to switch to cloth production: relative supply of cloth will increase.
 - Domestic consumers must pay a higher relative price of cloth to producers, and therefore will be more willing to switch to food consumption: relative demand for cloth will decrease.

Effects of an Export Subsidy (2 of 2)

- When the home country imposes an export subsidy, the terms of trade decrease and the welfare of the country decreases to the benefit of the foreign country.

Figure 6.10 Effects of a Cloth Subsidy on the Terms of Trade



An export subsidy on cloth has the opposite effects on relative supply and demand than the tariff on food. Relative supply of cloth for the world rises, while relative demand for the world falls. Home's terms of trade decline as the

relative price of cloth falls from $\left(\frac{P_C}{P_F}\right)^1$ to $\left(\frac{P_C}{P_F}\right)^2$.

Implications of Terms of Trade Effects: Who Gains and Who Loses? (1 of 4)

- The standard trade model predicts that
 - an import tariff by the home country can increase domestic welfare at the expense of the foreign country.
 - an export subsidy by the home country reduces domestic welfare to the benefit of the foreign country.

Implications of Terms of Trade Effects: Who Gains and Who Loses? (2 of 4)

- Additional effects of tariffs and subsidies that can occur in a world with many countries and many goods:
 - A foreign country may subsidize the export of a good that the United States also exports, which will reduce the price for the United States in world markets and decrease its terms of trade.
 - The EU subsidizes agricultural exports, which reduce the price that American farmers receive for their goods in world markets.
 - A foreign country may put a tariff on an imported good that the United States also imports, which will reduce the price for the United States in world markets and increase its terms of trade.

Implications of Terms of Trade Effects: Who Gains and Who Loses? (3 of 4)

- Export subsidies by foreign countries on goods that
 - the U.S. imports **reduce the world price of U.S. imports** and increase the terms of trade for the United States
 - the U.S. also exports **reduce the world price of U.S. exports** and decrease the terms of trade for the United States
- Import tariffs by foreign countries on goods that
 - the U.S. exports **reduce the world price of U.S. exports** and decrease the terms of trade for the United States
 - the U.S. also imports **reduce the world price of U.S. imports** and increase the terms of trade for the United States.

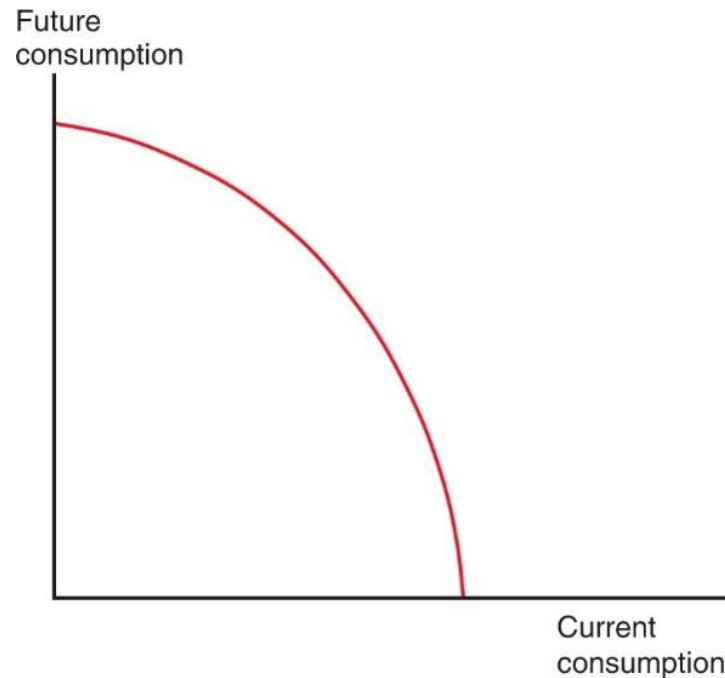
Implications of Terms of Trade Effects: Who Gains and Who Loses? (4 of 4)

- Export subsidies on a good **decrease the relative world price** of that good by increasing relative supply of that good and decreasing relative demand of that good.
- Import tariffs on a good **decrease the relative world price** of that good (and increase the relative world price of other goods) by increasing the relative supply of that good and decreasing the relative demand of that good.

International Borrowing and Lending (1 of 5)

- The standard trade model can be modified to analyze international borrowing and lending.
 - Two goods are current and future consumption (same good at different times), rather than different goods at the same time.
- Countries usually have different opportunities to invest to become able to produce more in the future.
- A special kind of production possibility frontier, an **intertemporal production possibility frontier**, depicts different possible combinations of current output and future output.

Figure 6.11 The Intertemporal Production Possibility Frontier



A country can trade current consumption for future consumption in the same way that it can produce more of one good by producing less of another.

International Borrowing and Lending (2 of 5)

- Suppose that Home has production possibilities biased toward current output, while Foreign has production possibilities biased toward future output.
 - Foreign has better opportunities to invest now to generate more output in the future.

International Borrowing and Lending (3 of 5)

- If you borrow 1 unit of output, you must repay principal + interest = $1 + r$ in the future, where r is the **real interest rate**.
- The price of future consumption relative to current

consumption is $\frac{1}{(1+r)}$.

- 1 unit of current consumption is worth $1 + r$ of future consumption,
 - so 1 unit of future consumption is worth $\frac{1}{(1+r)}$ units of current consumption.

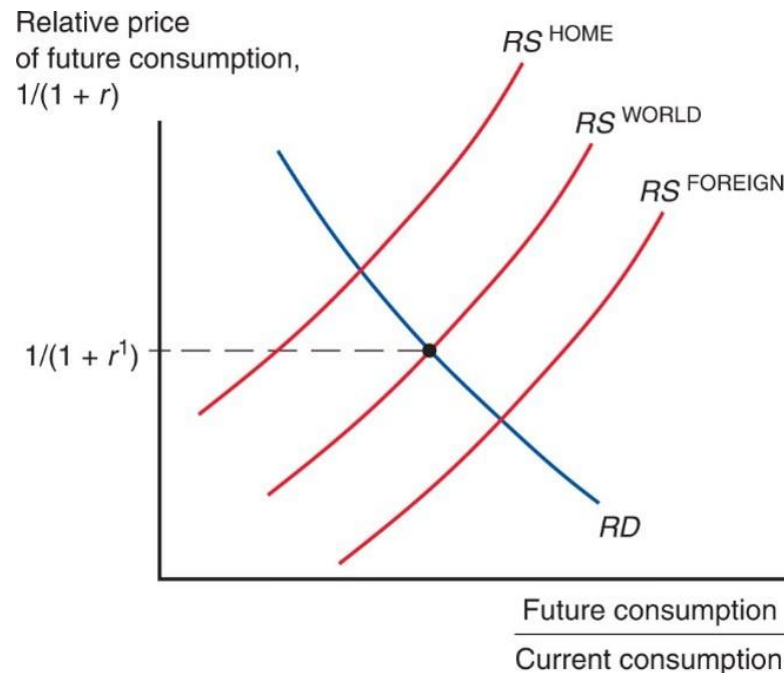
International Borrowing and Lending (4 of 5)

- Home exports current consumption and imports future consumption.
- Home lends to Foreign by consuming less than it produces now. Home able to consume more than produces in the future when Foreign pays back the loan.
- Foreign borrows to be able to consume more than produces now, and pays back the loan by consuming less than produces in the future.

International Borrowing and Lending (5 of 5)

- When international borrowing and lending are allowed, the world real interest rate is determined by the intersection of world relative demand and world relative supply.
 - The world real interest rate will be between the real interest rates that existed in the two country's prior to intertemporal trade.
 - The real interest rate rises in the country that lends (home) and falls in the country that borrows (foreign) due to intertemporal trade.

Figure 6.12 Equilibrium Interest Rate with Borrowing and Lending



Home, Foreign, and world supply of future consumption relative to current consumption. Home and Foreign have the same relative demand for future consumption, which is also the relative demand for the world. The equilibrium interest rate is determined by the intersection of world relative supply and demand.